ROOT CAUSE UNLOCKED

Unlock Your *Allergies*

A Functional Medicine Guide to Finding and Fixing the Root Cause

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Unlock Your Allergies

A Functional Medicine Guide to Finding and Fixing the Root Cause

BEFORE YOU START

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Chapter 1: Allergies Are Not the Problem

If you have been living with allergies, you know the cycle. Antihistamines in the morning, nasal sprays throughout the day, and a constant low-grade misery that nobody around you quite understands because from the outside, it just looks like a runny nose. What the conventional model rarely addresses is why your immune system became hypersensitive in the first place.

Allergies are not a Zyrtec deficiency. They are a signal that the immune system has lost its ability to tolerate things it should not be reacting to. The question functional medicine asks is not how to suppress that reaction but what broke the immune system's tolerance mechanisms in the first place.

What Is Actually Happening

An allergic response is an immune system overreaction driven primarily by IgE antibodies and mast cells. When the immune system encounters a trigger, whether it is pollen, pet dander, dust, or a food, it releases histamine and other inflammatory chemicals.[1] These chemicals cause the symptoms you feel: runny nose, itchy eyes, hives, sneezing, swelling, and in severe cases, anaphylaxis.

But why does one person sneeze their way through spring while another has no response to the same pollen? The difference lies in the condition of the terrain. A gut that has lost its barrier integrity, a liver that is congested and struggling to clear histamine, an adrenal system that is depleted and cannot produce adequate cortisol to regulate inflammation, a microbiome that is dysbiotic and constantly stimulating the immune system. These are not separate issues. They are the root causes underneath the allergic phenotype.

> ***The Systems Behind Allergies***
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> - *Gut integrity and microbiome health: the figure usually quoted is that around 70% of immune tissue sits in the gut, which is a rough approximation rather than a measured constant, but the direction it points is not in dispute*
> - *Histamine metabolism: two enzyme pathways (DAO and HNMT) clear histamine; genetic variants in either can produce treatment-resistant histamine intolerance regardless of dietary changes*
> - *Adrenal function: cortisol is the body's natural anti-inflammatory hormone*
> - *Toxic burden: mold, heavy metals, and chemicals amplify mast cell reactivity*
> - *Nutrient status: vitamin C, vitamin D, quercetin, and magnesium are all essential to immune regulation*
> - *Food sensitivities: often a hidden driver that keeps the immune system chronically activated*

Why Antihistamines Are Not Enough

Antihistamines work by blocking histamine receptors, which reduces symptoms in the short term. This can be genuinely useful and there is nothing wrong with getting relief. The problem is that blocking the receptor does not address why histamine is being produced in excess, why the body is not clearing it efficiently, or why the immune system is reacting in the first place.

Long-term antihistamine use is also argued about more than the marketing suggests. Sedating formulations have a documented cognitive cost, and the further claims made in wellness circles, nutrient depletion and rebound worsening of histamine intolerance, are proposed mechanisms rather than settled ones. I raise them so you know the discussion exists, not to talk you off a medication that is helping you.

Allergen immunotherapy, the allergy shots and sublingual drops managed by an allergist, works through a different mechanism, gradually desensitizing the immune system. It is more aligned with a root cause approach than daily suppression, and it is not something I prescribe. It is included here so you know it exists and can raise it with the physician managing that part of your care. What it still does not address is the gut, the liver, or the toxic burden that lowered your tolerance threshold in the first place, which is the work this book is about.

The Root Health Approach

The Two Histamine-Clearing Enzyme Pathways

This section is worth understanding in detail because it explains why some people do everything right, clean diet, quercetin, vitamin C, gut work, and still cannot get their histamine burden under control. The answer is often genetic.

The body clears histamine through two distinct enzymatic pathways. DAO (Diamine Oxidase), encoded by the AOC1 gene, is the primary enzyme for breaking down dietary and extracellular histamine in the gut.[2] It degrades histamine before it is absorbed, acting as the first line of clearance for everything that comes in through food and the gut environment. HNMT (Histamine N-Methyltransferase) handles intracellular histamine clearance, primarily in the tissues. It requires SAM (S-adenosylmethionine) as a methyl donor, which means your methylation capacity directly determines how efficiently this pathway operates.

Both genes have well-characterized SNP variants that reduce enzyme activity. AOC1 SNPs can reduce DAO enzyme function significantly, meaning dietary histamine that should be degraded in the gut passes into circulation and accumulates. The most clinically significant HNMT variant, Thr105Ile (C314T), reduces intracellular histamine clearance roughly threefold in individuals who carry it.[3] When both pathways are compromised simultaneously, histamine accumulates rapidly regardless of how low the dietary intake is.

This is also why Methyl-B12 is not simply generic methylation support in this protocol. HNMT requires methylation to function. Supporting methylation through Methyl-B12 and methylated B vitamins directly supports intracellular histamine clearance through the HNMT pathway. For patients with MTHFR variants reducing methylation capacity, addressing methylation is not optional, it is a direct intervention in histamine metabolism.

DAO enzyme activity can be measured functionally through the Advanced Intestinal Barrier Assessment at Precision Point Diagnostics. This is a serum test that measures what the enzyme is actually doing, not just which genetic variant someone carries. A person can have a low-risk AOC1 genotype and still have functionally low DAO activity due to nutrient cofactor depletion. DAO requires vitamin B6, copper, and vitamin C to function. Conversely, testing only genetics without function misses the patients whose enzyme is structurally intact but nutritionally disabled. Both pieces of information matter clinically.

At Root Health, we approach allergies as we do every chronic condition: by identifying the underlying dysfunction driving the symptom pattern. This means comprehensive testing to assess gut integrity, food sensitivities, histamine-processing capacity, nutrient status, and toxic burden. It means a drainage-first approach that opens the body's clearance pathways before loading in targeted interventions. And it means addressing the whole system rather than chasing individual symptoms.

This guide walks you through the dietary foundations, the laboratory tests that provide the most actionable data, the supplement protocols we use clinically, and the lifestyle tools that accelerate recovery. Whether you are working through these steps on your own or alongside a practitioner, the information here is designed to move you from managing symptoms to actually resolving the underlying problem.

Chapter 2: The Allergy Diet

Diet is not a peripheral consideration in allergies. It is central. What you eat either loads the immune system further or gives it the conditions it needs to calm down and rebuild tolerance. Two mechanisms matter most: food sensitivities that keep the immune system chronically activated, and high dietary histamine loads that overwhelm the body's already-compromised clearing capacity.

Step One: Remove the Triggers

High-Histamine Foods

If your histamine bucket is already full from environmental allergens, the last thing you need is to fill it further through diet. High-histamine foods are processed, aged, fermented, or smoked. They either contain high levels of histamine directly or trigger mast cells to release more.

- Aged cheeses (parmesan, blue cheese, cheddar, brie)
- Wine, beer, and champagne
- Fermented foods (sauerkraut, kimchi, kombucha, miso, soy sauce)
- Smoked and cured meats (salami, bacon, deli meats)
- Shellfish and canned fish
- Vinegar and vinegar-containing condiments
- Processed and packaged foods
- Tomatoes, eggplant, spinach, avocado (histamine liberators)
- Alcohol in any form

This list is not meant to be permanent. The goal is a 4 to 6 week low-histamine period to reduce the total burden on your system while you address the underlying causes. As gut integrity improves and the liver's histamine-clearing capacity is supported, most people can reintroduce these foods without issue.

Gluten and Dairy

These are the two most common food sensitivities in patients with environmental allergies and chronic immune activation. Gluten triggers zonulin release, which opens tight junctions in the intestinal wall and worsens gut permeability.[4] Dairy contains casein proteins that cross-react with other allergens and stimulate mucus production in sensitive individuals. Removing both for 6 to 8 weeks is not a permanent lifestyle sentence but a diagnostic intervention that often produces dramatic reductions in total allergic load.

Artificial Additives and Food Dyes

Artificial colors, preservatives, and flavor enhancers are known mast cell triggers. Tartrazine (Yellow 5), sodium benzoate, and sulfites have documented associations with allergic reactions and histamine release. If it comes in a package with an ingredient list that reads like a chemistry textbook, it is contributing to your allergic burden.

Step Two: Build the Anti-Allergy Plate

Quercetin-Rich Foods

Quercetin is a natural flavonoid that functions as a mast cell stabilizer, reducing histamine release before it starts.[5] It is one of the most clinically useful natural compounds for allergy management and the foods highest in it should anchor the diet.

- Onions (especially red onions, raw)
- Capers (the highest natural quercetin source by weight)
- Apples with the skin
- Blueberries, blackberries, and dark cherries
- Broccoli, kale, and leafy greens
- Green tea

Omega-3-Rich Foods

Omega-3 fatty acids reduce systemic inflammation and blunt the inflammatory signaling that amplifies allergic responses.[6] Cold-water fatty fish, ground flaxseed, chia seeds, and walnuts should appear in the diet multiple times per week.

Vitamin C-Rich Foods

Vitamin C is a natural antihistamine and antioxidant that supports the adrenal glands and assists histamine clearance. Fresh bell peppers, citrus (tolerated in low-histamine phase), kiwi, strawberries, and broccoli are reliable sources. Note that some citrus and strawberries are histamine liberators in sensitive individuals, so assess tolerance individually.

Prebiotic Fiber and Gut-Healing Foods

The microbiome directly regulates immune reactivity. Feeding beneficial bacteria with prebiotic fiber and supporting gut barrier integrity are foundational steps that reduce the antigenic load driving immune hypersensitivity.

- Garlic and leeks (prebiotic)
- Asparagus and artichokes

- Oats (if tolerated and certified gluten-free)
- Bone broth (gut-healing collagen and glycine)
- Cooked and cooled potatoes and rice (resistant starch)

Hydration and Liver Support

Liver support belongs in this chapter, and the reason usually given for it is wrong, so here is the accurate version. Diamine oxidase is not primarily a liver enzyme. It is secreted and concentrated in the tips of the villi in the small intestine, along with kidney and placenta, and it handles histamine that arrives from food and histamine released into the space between cells. The liver's contribution runs through a different enzyme, histamine N-methyltransferase, which works inside cells and is the route that serves the brain and airway. Two enzymes, two compartments, and they are not interchangeable. What follows practically is that gut lining health, not liver congestion, is the dominant lever on the enzyme that clears dietary histamine, which is why the gut work in this program sits upstream of anything else. Hydration, cruciferous vegetables, beets and dandelion greens remain reasonable general support for hepatic conjugation and bile flow, and I keep them here on that basis rather than on a diamine oxidase claim that the physiology does not support.

> ***The Anti-Allergy Plate: Quick Start***
>
> - *Half your plate: non-starchy vegetables (leafy greens, broccoli, zucchini, asparagus, onions)*
> - *A quarter of your plate: clean protein (grass-fed beef, pastured chicken, wild salmon, eggs if tolerated)*
> - *A quarter of your plate: complex carbohydrates (sweet potato, rice, quinoa)*
> - *Add daily: fresh herbs, raw onion, apple slices, blueberries*
> - *Drink: filtered water with lemon, green tea, bone broth*
> - *Avoid for 6 weeks: gluten, dairy, fermented foods, alcohol, aged or smoked proteins, artificial additives*

Chapter 3: Lab Testing for Allergies

Symptoms tell you something is wrong. Lab testing tells you what and where. Guessing at the root cause of allergies without objective data leads to protocols that miss the mark. The tests below are the most clinically informative for identifying the root systems driving allergic hypersensitivity.

Test	What It Measures	Why It Matters for Allergies
Food Zoomer (Vibrant)	IgG and IgA antibodies to 180+ foods; identifies delayed sensitivities	Hidden food sensitivities chronically activate the immune system and keep total allergic load high, amplifying reactions to environmental triggers
Wheat Zoomer (Vibrant)	Wheat and gluten antibodies, zonulin, intestinal permeability markers, celiac genetics	Screens gluten reactivity and celiac genetics, and reads several barrier markers together. **Read the barrier markers as a pattern, not as confirmation.** Serum zonulin measured by commercial kit has been shown not to correlate with the physiologic lactulose-mannitol measure of permeability,[12] so an elevated zonulin is suggestive rather than confirmatory. The IgA anti-actin component is the best validated marker on the panel, though its validation is specifically for celiac disease with villous atrophy, where it reached 80% sensitivity and 85% specificity[13]
Gut Zoomer (Vibrant)	300+ microbiome species, secretory IgA, permeability, inflammation, parasites, yeast	Gut dysbiosis and low secretory IgA are primary drivers of immune hypersensitivity. The microbiome directly regulates whether the immune system mounts tolerant or reactive responses
Micronutrient Panel (Vibrant)	Intracellular levels of vitamin C, D, zinc, magnesium, B vitamins, quercetin, antioxidants	Vitamin C, D, and zinc deficiencies directly impair immune regulation and histamine clearance. You cannot supplement effectively without knowing what is actually depleted at the cellular level
Total Tox Burden (Vibrant)	Heavy metals, mycotoxins, and environmental chemicals in one panel	Mold toxins and heavy metals are major amplifiers of mast cell reactivity. Chronic toxic burden is an underappreciated driver of treatment-resistant allergies
Organic Acids Test (Vibrant)	Neurotransmitter metabolites, microbial overgrowth markers, mitochondrial function, histamine-related metabolites	Candida and certain bacterial overgrowths produce histamine directly. OAT reveals gut dysbiosis patterns that standard stool tests miss, including histamine-producing organisms
Advanced Intestinal Barrier Assessment (Precision Point Diagnostics)	Functional serum DAO enzyme activity; also includes intestinal permeability markers (occludin antibodies, zonulin antibodies, actomyosin antibodies, LPS antibodies)	Measures what DAO is actually doing, not just what gene variant is present. Low functional DAO is consistent with histamine intolerance as a primary driver rather than proof of it, and it is read alongside the symptom pattern. Nutrient cofactor depletion (B6, copper, vitamin C) can produce low DAO activity even without a genetic variant. Essential when histamine symptoms persist despite dietary compliance.

Test	What It Measures	Why It Matters for Allergies
Genetic Testing: AOC1 and HNMT SNPs (Genova DetoxiGenomic or standalone SNP panel)	AOC1 gene variants reducing DAO enzyme activity; HNMT Thr105Ile (C314T) variant reducing intracellular histamine methylation; MTHFR variants impairing the methylation capacity HNMT depends on	Identifies whether histamine intolerance has a genetic foundation. AOC1 variants explain why dietary changes alone may not resolve symptoms. HNMT variants explain why Methyl-B12 is mechanistically required, not optional, for intracellular histamine clearance.[2,3] Run alongside functional DAO testing for the complete picture.

What These Tests Reveal

Food Sensitivities and the Total Allergic Load Concept

The total allergic load model holds that the immune system has a threshold for reactivity. When multiple immune stressors, food sensitivities, environmental allergens, gut dysbiosis, toxic burden, combine, they push the system over that threshold into symptomatic reactivity. The Food Zoomer identifies IgG and IgA-mediated food sensitivities, which are delayed reactions that can occur hours or days after exposure, making them invisible without testing.

Many patients find that removing their primary food sensitivities reduces their environmental allergy symptoms substantially, even before addressing the environmental triggers directly. The food sensitivity load was the variable that was pushing them over threshold.

Gut Permeability and Immune Education

The gut-associated lymphoid tissue (GALT) is where the immune system learns what to react to and what to tolerate. When the intestinal barrier is compromised, from gluten, stress, medications, dysbiosis, or toxic exposure, partially digested food proteins cross into the bloodstream and are treated as threats. This is one of the primary mechanisms through which food sensitivities develop and through which environmental allergies worsen over time.

Secretory IgA, measured on the Gut Zoomer, is the front-line mucosal defense antibody. Low secretory IgA indicates a depleted mucosal immune system that is poorly positioned to manage allergen exposure. Elevated secretory IgA indicates active immune upregulation in the gut, often from food sensitivities, infections, or ongoing permeability.[7]

Histamine-Producing Organisms

Certain bacteria and yeast species produce histamine directly as a metabolic byproduct. When these organisms are present in significant quantities, they contribute to the total histamine load from within, independent of dietary intake. The Organic Acids Test and Gut Zoomer identify these patterns, making targeted treatment of dysbiosis a direct intervention against histamine excess rather than a tangential one.

DAO and HNMT: The Two Histamine-Clearing Pathways

Understanding why histamine accumulates matters as much as understanding where it comes from. The body has two primary enzymatic pathways for clearing histamine, and dysfunction in either, whether from genetic variants, nutrient depletion, or gut damage, produces chronic histamine excess regardless of how strictly dietary intake is managed.

DAO (Diamine Oxidase) is the primary enzyme for degrading dietary and extracellular histamine in the gut.[2] It works in the intestinal lining before histamine is absorbed, functioning as the first line of defense against histamine load from food and the gut environment. DAO activity requires vitamin B6, copper, and vitamin C as cofactors. A person with adequate AOC1 genetics can still have functionally low DAO if these cofactors are depleted, which is extremely common in people with chronic gut inflammation or poor absorption. Gut damage from dysbiosis, gluten exposure, or chronic inflammation also reduces the number of DAO-producing intestinal cells, compounding the problem.

Functional serum DAO activity through Precision Point Diagnostics Advanced Intestinal Barrier Assessment is the most clinically useful way to assess this pathway because it measures what the enzyme is actually doing in the patient, not just which genetic variant they carry. It also includes intestinal permeability markers: occludin antibodies, zonulin antibodies, actomyosin antibodies, and LPS antibodies, making it a broad gut barrier and histamine-processing panel in a single draw.

A note on how to read those barrier markers, because they are not interchangeable. They measure genuinely different things and their evidence differs a great deal. IgA anti-actin is the best validated, with 80% sensitivity and 85% specificity and an area under the curve of 0.873, though that validation is for celiac disease with villous atrophy rather than permeability in general.[13] LPS antibodies reflect bacterial translocation, and notably they track separately from markers of intestinal damage, so they answer a different question rather than a confirming one.[14] Serum zonulin by commercial kit failed to correlate with the physiologic lactulose-mannitol measure in one study of healthy adults.[12] I could not locate comparable human validation for occludin antibodies. Read the panel as a pattern across markers, and be cautious about treating any single one as proof.

HNMT (Histamine N-Methyltransferase) handles intracellular histamine clearance throughout the body's tissues. Unlike DAO, HNMT requires SAM (S-adenosylmethionine) as a methyl donor, which means methylation capacity directly gates how efficiently histamine is cleared at the cellular level. The Thr105Ile (C314T) HNMT variant reduces enzyme activity roughly threefold.[3] Patients carrying this variant alongside MTHFR polymorphisms that reduce methylation capacity face a compounded histamine-clearing deficit: reduced HNMT enzyme activity combined with insufficient SAM to drive what activity remains. This is the clinical rationale for prioritizing Methyl-B12 and methylated B vitamins in this protocol, they are not generic wellness supplements here, they are mechanistically tied to intracellular histamine clearance.

When a patient has tried dietary changes, taken quercetin and vitamin C, and still cannot get histamine symptoms under control, the answer is frequently here: low functional DAO, an HNMT variant, poor methylation, or some combination of all three. Testing makes this visible and actionable rather than a guessing game.

Mold and Toxic Burden

Mycotoxins from mold exposure are proposed as potent activators of mast cells, and mold history is one of the most commonly missed items in treatment-resistant allergy. If you have been in a water-damaged building, live in a humid climate, or have a mold exposure history, this belongs in the workup rather than at the bottom of the list.

Be clear about what the panel does. The Total Tox Burden measures urinary mycotoxins alongside heavy metals and environmental chemicals, and a urinary panel documents recent exposure. It does not measure what is stored in tissue, and it does not by itself establish that mold is driving your allergy picture. I read it as one line of evidence next to the exposure history, the building, and the rest of the labs. Where the panel and the history agree, that is where I start.

Chapter 4: Supplementation Protocols

Supplements for allergies serve several distinct functions: stabilizing mast cells to reduce histamine release, supporting the clearance of histamine already in circulation, healing gut integrity to reduce the antigenic load entering the system, modulating immune reactivity at the root level, and providing the raw nutritional materials the body needs to regulate its own inflammatory response.

Everything below is a single ingredient or a standardized botanical. Nothing here is practitioner-exclusive, and nothing requires ordering through this office.

What you are buying here, stated plainly. Most of this table rests on laboratory evidence and long clinical use, not on human outcome trials in allergic rhinitis. That is true of quercetin, stinging nettle, bromelain, and the DAO enzyme alike, and it is what happens to compounds nobody can patent: a large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. So what follows is reasoned clinical practice held to an endpoint, not trial-proven certainty, and I would rather you know which one you are following. Two consequences worth acting on. Put your money into the terrain work first, the diet in Chapter 2 and the gut repair, because that is where the leverage is. Then hold this protocol to a defined result rather than running it indefinitely.

The supplement protocol

The following are the best evidence-supported options. Form matters more than brand, so each entry names the form that was studied.

Supplement	Form to look for	Dose	Clinical Purpose
Quercetin	Quercetin Phytosome; the phytosome form is substantially more bioavailable than standard quercetin	500-1000mg twice daily, 20 min before meals	Mast cell stabilizer. Be aware what the evidence is: quercetin outperformed cromolyn at blocking cytokine release from human mast cells in the laboratory, and helped contact dermatitis and photosensitivity in people.[5] That is mechanism plus two skin conditions, not a trial in allergic rhinitis. Nobody has funded the rhinitis trial, because quercetin is not patentable. I still use it, at the phytosome form and the doses listed, because the mast cell mechanism is the one we are actually trying to move and the safety margin is wide. I hold it to eight weeks against your own symptom log, and if the log has not moved, it comes out
Vitamin C	Buffered ascorbate or liposomal vitamin C; avoid plain ascorbic acid if you are stomach-sensitive	2 g per day, divided	Dosed to the histamine target: 2 g is the oral dose that has actually been studied against histamine, and higher intakes have declining absorption with the surplus excreted
Stinging Nettle Leaf	Freeze-dried whole leaf; freeze-drying preserves constituents that heat processing destroys	300-600mg twice daily at onset of symptoms	Natural antihistamine via DAO enzyme support and mast cell modulation; well-documented in research for seasonal allergy relief[8]
Bromelain	Proteolytic enzyme from pineapple stem; take away from food for the anti-inflammatory effect	500-1000mg twice daily between meals	Reduces mucosal inflammation; enhances quercetin absorption; reduces nasal congestion and sinusitis from allergic inflammation
Omega-3 (EPA/DHA)	Triglyceride-form fish oil; avoid ethyl ester forms	2-3g combined EPA/DHA daily	Reduces leukotriene production and systemic inflammation that amplifies allergic reactions[6]
Vitamin D3 + K2	D3 as cholecalciferol with K2 as MK-7; test levels first and dose into the functional range of 40 to 80 ng/mL	Dose to your measured level rather than to a fixed number	Immune modulator. The supporting systematic review and meta-analysis studied supplementation and allergic disease **in childhood**,[9] so it is reasonable grounds for correcting a documented deficiency in adults and not evidence of an adult allergy treatment effect. That is exactly how I use it: I correct a measured low result because deficiency is common, cheap to fix,

Supplement	Form to look for	Dose	Clinical Purpose
			and defensible on general grounds, and I do not count on it to treat your allergies. Test rather than guess. K2 ensures proper calcium routing
Magnesium Glycinate	Glycinate form only; avoid oxide forms	300-400mg at bedtime	Mast cell membrane stabilizer; magnesium deficiency increases mast cell reactivity and histamine release
DAO Enzyme Supplement	Bovine or porcine kidney-derived DAO	1-2 capsules before high-histamine meals	Breaks down dietary histamine before absorption, which is a sound mechanism. Worth knowing that the randomized placebo-controlled trial of a low-histamine diet plus DAO supplementation has been **designed and published as a protocol, and its results are not yet in**.[10] So this is mechanism and clinical experience, not a demonstrated outcome. Reasonable to trial, honest to say the proof is pending
Probiotics (Multi-strain)	Lactobacillus rhamnosus GG, L. reuteri, and Bifidobacterium species	25-50 billion CFU daily with food	Strain selection matters more than total count here, and the effects reported in the probiotic literature are strain specific rather than general to the species
NAC (N-Acetyl L-Cysteine)	Standard NAC	600mg twice daily away from meals	Glutathione precursor that supports liver detoxification and histamine clearance; also thins mucus secretions in sinuses and airways
Transfer Factor	Colostrum-derived transfer factor	As directed on label	Colostrum-derived immune signaling molecules that help educate and modulate immune response; overlapping territory with colostrum

> ***A note on supplement quality***
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> *The supplement industry is largely unregulated, and potency, purity, and bioavailability vary enormously between products. Look for independent third-party testing (NSF, USP Verified, or equivalent), and match the form named in the table rather than the name on the front of the bottle. Generic quercetin, omega-3, and probiotic products in particular are frequently under-dosed or in forms that are poorly absorbed.*
>
> *If you would rather not evaluate all of that yourself, I keep a screened dispensary through Fullscript, which ships directly to you. I hold no physical inventory. It is screened on the standards above and organized by the categories in this chapter.*
>
> ***Root Health dispensary: us.fullscript.com/welcome/drseay***

Disclosure: Root Health earns a portion of dispensary purchases. Nothing here was included, excluded, ranked, or dosed based on what it earns, and you can buy any of it anywhere. Several of the most useful items in this book, including nasal irrigation and air quality changes, earn us nothing at all.

Chapter 5: Therapeutic Modalities

Supplements and diet address the internal biochemistry. These modalities address the physical and environmental inputs that directly affect mast cell reactivity, mucosal health, lymphatic clearance, and immune regulation. Used consistently, they accelerate and deepen the results of the nutritional protocol.

Nasal Irrigation

Daily nasal rinsing with a saline solution removes pollen, dust, mold spores, and other allergens from the nasal passages before the immune system has a chance to react to them. It also reduces mucosal inflammation and improves ciliary function, which is the mechanical clearance system of the upper respiratory tract. Research supports nasal irrigation as a meaningful adjunct to allergy management, with studies showing reductions in symptom scores and medication use.[11]

The NeilMed sinus rinse system or a neti pot with non-iodized salt in distilled or previously boiled water is the standard approach. Once daily during high-pollen periods, or twice daily during acute flares. Do not use tap water.

Air Quality Management

Environmental allergen burden matters enormously. The average American spends over 90 percent of their time indoors, and indoor air quality is frequently worse than outdoor air due to dust mites, pet dander, mold spores, and chemical off-gassing from building materials and cleaning products.

- HEPA air purifier in the bedroom and main living space (Austin Air or IQAir are clinical-grade options)
- MERV-13 or higher furnace filters replaced every 30-60 days
- Mattress and pillow encasements to reduce dust mite exposure
- Reduce indoor humidity to below 50% to prevent mold growth
- Remove carpeting where possible; vacuum remaining carpet with a HEPA filter vacuum
- Eliminate synthetic fragrances, conventional cleaning products, and plug-in air fresheners (direct mast cell triggers)
- Open windows during low-pollen hours (after rain, late afternoon) and keep them closed during peak pollen times

Infrared Sauna

Sauna shows up in most terrain-based protocols, and the mechanisms usually given for it are proposals rather than settled findings. Sweating as a route of toxin elimination is real for some compounds and negligible for others, and nobody has funded a trial asking whether it changes allergy symptoms.

Parasympathetic activation and improved lymphatic flow are the other two mechanisms offered, and both are plausible without being demonstrated in this population.

Here is how I actually use it, since you paid for a position rather than a shrug. Sauna is a supportive tool while the exposure work, the diet, and the gut repair do the heavy lifting. It has a place, and it is not the plan by itself. If the eight-week trial below does not move your log, it is not the piece that was missing.

If you use one: 15-20 minutes at low temperature (120-130 degrees Fahrenheit) two to three times per week, building gradually toward 30-45 minutes as tolerance allows. Hydrate and replace electrolytes. Avoid it in pregnancy, and clear it with a clinician first if you have cardiovascular disease. Start very low and slow if mold toxicity is suspected, and treat worsening as a signal to back off rather than as proof it is working.

Vagal Nerve Support

The vagus nerve is the primary regulator of the parasympathetic nervous system and has direct anti-inflammatory effects via the cholinergic anti-inflammatory pathway. Chronic stress and sympathetic dominance, which is the state most people with allergies live in, suppresses vagal tone and increases inflammatory signaling including mast cell reactivity. Supporting vagal tone is therefore a direct intervention into the inflammatory architecture of allergy.

- Diaphragmatic breathing: 5-count inhale through the nose, 7-count exhale through the mouth, 10 minutes daily
- Cold water face immersion or cold shower finish for acute vagal activation
- Humming, gargling, and singing activate the vagal nerve through pharyngeal muscle stimulation
- Devices: Apollo Neuro (wearable vibration), Pulsetto, or Nurosym for sustained vagal stimulation

Lymphatic Support

The lymphatic system carries immune cells and removes cellular debris and allergen-immune complexes from tissues. When lymphatic flow is sluggish, immune reactants accumulate in tissues and the body struggles to clear the aftermath of each allergic reaction efficiently.

- Rebounding (mini-trampoline): 10-15 minutes daily; the vertical movement is the most effective mechanical pump for lymphatic flow
- Dry brushing: 5 minutes before showering, brushing toward the heart in long strokes
- Lymphatic massage: professional or self-administered lymphatic drainage around the neck, armpits, and groin where nodes concentrate
- Walking: even 20-30 minutes of brisk walking significantly improves lymphatic circulation

Exercise

Moderate aerobic exercise has documented anti-inflammatory effects and reduces overall histamine burden over time. Exercise increases cortisol in a physiologically appropriate way, temporarily raising the threshold for allergic reactivity. It also improves lymphatic flow, reduces stress hormones, and supports healthy microbiome diversity, all of which contribute to lower baseline allergic reactivity.

The caveat is exercise-induced bronchoconstriction, which is a real phenomenon in exercise-sensitive individuals with asthma or severe respiratory allergies. Start with low-intensity exercise like walking or swimming and build gradually. Swimming in outdoor bodies of water or well-maintained pools is often better tolerated than dry, dusty outdoor environments during high-pollen seasons.

Target 30-45 minutes of moderate aerobic activity four to five times per week. Morning exercise before peak pollen release is preferable for those with strong environmental sensitivities.

Chapter 6: Working with Root Health

Everything in this guide can move the needle. But there is a ceiling on what self-directed protocols can accomplish without objective data, professional interpretation, and a clinician who can adjust the protocol based on what your labs actually show. Most people with complex or treatment-resistant allergies have multiple root causes operating simultaneously, and addressing only one or two of them explains why the results often plateau.

The M4 Method

At Root Health, I practice through the M4 Method, a four-pillar framework for comprehensive functional medicine care:

- Movement: corrective exercise and restored mobility, which support nervous system function and the structural integrity that lymphatic and immune regulation depend on
- Manipulation: chiropractic adjustment, used to reduce sympathetic dominance and support vagal tone. The anti-inflammatory effect of that shift is a proposed mechanism rather than a settled one, and I am telling you which it is
- Medicinal Nutrition: targeted supplement protocols built on objective laboratory findings and calibrated to the deficiencies and dysfunction patterns your testing actually shows
- Metabolic Connection: the upstream drivers, gut health, detoxification capacity, hormone balance, and cellular energy production, that decide whether any other intervention holds

The Assessment Process

The first step is always finding out what is actually happening, and I build that from three things: a detailed health history, a physical examination, and laboratory testing. Nothing else. The history tells me what changed and when, the exam tells me what your body is doing right now, and the labs give me numbers I can retest later to find out whether I was right.

For allergy patients specifically, I am looking for convergence. Where do the history, the exam, and the labs all point at the same system? That agreement is where I start, because a finding that shows up in only one of the three is usually noise, and a finding that shows up in all three is usually the thing worth your first ninety days.

The Three Phases of Care

Phase 1: Drainage and Foundation (Months 1-2)

Before any targeted allergy intervention, we establish drainage. This means opening the lymphatic, liver, kidney, and bowel pathways so that the immune and detoxification work that follows has somewhere to go. In allergy cases, this phase typically includes gut restoration, liver support, and initial histamine burden reduction through diet and foundational supplements. Jumping straight to aggressive immune modulation or detoxification without drainage in place is where I see people feel worse instead of better, and that failure mode is common enough to be worth sequencing around.

One caution about how that worsening gets explained, because it matters. You will hear it called a Herxheimer or die-off reaction, and you will hear that feeling worse proves the protocol is working. If feeling better proves it works and feeling worse also proves it works, the claim cannot be tested, and an untestable claim is a poor reason to keep paying. So I treat worsening as information rather than as a milestone. Sometimes it does pass in a few days. Sometimes it means the dose is too high, the drainage groundwork was skipped, or this is the wrong protocol for you. Significant worsening is a reason to reassess with me, not a badge. And any sudden breathing difficulty, throat tightness, or facial swelling is not die-off at all, it is an emergency.

Phase 2: Targeted Intervention (Months 2-4)

With drainage established and the gut beginning to heal, we address the specific root causes identified in testing: food sensitivity elimination and gut repair, targeted dysbiosis treatment if histamine-producing organisms are present, toxic burden clearance if mycotoxins or heavy metals are elevated, adrenal rehabilitation, and immune modulation through the supplement protocol in Chapter 4. This phase is where the most significant clinical shifts happen.

Phase 3: Restoration and Maintenance

As acute imbalances resolve, we shift toward rebuilding resilience, restoring microbiome diversity, sustaining gut barrier integrity, optimizing nutrient status, and establishing the lifestyle practices that prevent recurrence. For most allergy patients, this is also the phase where environmental tolerance begins to genuinely expand and the total allergic load concept starts working in their favor rather than against them.

> ***The Root Discovery Session***
>
> *The entry point for working with Root Health is the Root Discovery Session, a comprehensive consultation covering your complete health history, current symptoms, assessment findings, and an initial protocol direction.*
>
> *The Root Discovery Session is free, and there is no obligation.*
>
> *To schedule: rootcauseunlocked.com/discovery, or call the office at (919) 662-0520.*
>
> *Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.*

> *This is not a sales call. It is a clinical consultation designed to determine whether and how I can help you and what that process would look like.*

Chapter 7: Your 90-Day Allergy Action Plan

The steps below are sequenced for a reason. Drainage first, then gut foundation, then targeted intervention, then immune modulation and detoxification. Jumping ahead produces slower results and more side effects. Follow the sequence.

Weeks 1 to 4: Foundation and Drainage

> ***Weeks 1-4 Priorities***
>
> - *Remove gluten, dairy, alcohol, fermented foods, and artificial additives completely*
> - *Begin low-histamine eating protocol from Chapter 2*
> - *Start foundational supplements: quercetin phytosome, vitamin C, omega-3, vitamin D3/K2, magnesium glycinate*
> - *Begin daily nasal irrigation (once daily, increasing to twice daily if symptomatic)*
> - *Assess your air quality: get a HEPA filter for the bedroom at minimum*
> - *Start diaphragmatic breathing practice: 10 minutes morning and evening*
> - *Begin walking 20-30 minutes daily*
> - *Order labs if not yet done: Food Zoomer and Gut Zoomer are the highest priority starting points*
> - *Begin drainage support: electrolytes through the day, methylated B12, and adaptogenic adrenal support*

Weeks 5 to 8: Gut Repair and Targeted Histamine Support

> ***Weeks 5-8 Priorities***
>
> - *Add colostrum or a colostrum-derived transfer factor for gut-immune barrier support*
> - *Add multi-strain probiotic (25-50 billion CFU) with food*
> - *Add DAO enzyme supplement before high-histamine meals if histamine intolerance is suspected*[10]
> - *Add NAC 600mg twice daily for liver and glutathione support*
> - *Add a digestive enzyme supplement with meals, ideally one including ox bile and betaine HCl*
> - *Begin infrared sauna 2-3 times per week at low temperature*
> - *Add rebounding 10-15 minutes daily for lymphatic support*
> - *Review lab results with practitioner and adjust protocol to findings*

> - *Begin elimination of specific foods identified on Food Zoomer*

Weeks 9 to 12: Immune Modulation and Toxic Burden

> ***Weeks 9-12 Priorities***
>
> - *Increase probiotic diversity, or add a spore-based probiotic*
> - *Add pancreatic enzyme support if digestive insufficiency is indicated*
> - *Begin addressing toxic burden if mycotoxins or heavy metals are elevated on Total Tox Burden*
> - *Increase infrared sauna frequency to 4 times per week if tolerated*
> - *Reassess symptom burden against your week 1 log, not against a number I gave you. Your own baseline is the only honest comparison, and the log from Chapter 7 is what makes it visible*
> - *Consider stinging nettle and bromelain during peak allergy season as acute support*
> - *Begin planning maintenance protocol with practitioner for post-acute phase*
> - *Schedule retest of primary labs at 6 months to document objective progress*

Tracking Your Progress

Keep a simple daily symptom log for the first 90 days: rate your nasal symptoms, eye symptoms, skin symptoms, energy, and sleep on a 1 to 10 scale each morning. This gives you objective data on your trajectory and helps identify which interventions are producing the most movement. It also gives your practitioner critical information for protocol adjustments.

In my clinical experience the dietary and foundational supplement work is where people notice something first, usually inside the first month or two, and the deeper change, the kind where you get through a season without reaching for antihistamines, takes longer and depends on what the testing actually found. I am not going to put a percentage on your outcome, because I have not examined you and any number I offered would be invented. Your log against your own baseline is the honest measure, and the retest at six months is the objective one.

APPENDIX

Quick Reference

Protocol Summary: Allergies

Support	Target	Typical range
Quercetin, phytosome form	Mast cell stabilization	500 to 1,000 mg twice daily, before meals
Vitamin C, buffered or liposomal	Histamine support and antioxidant	2 g per day, divided
Stinging nettle, freeze-dried leaf	Seasonal symptom support	300 to 600 mg twice daily at symptom onset
Bromelain	Mucosal inflammation, absorption of quercetin	500 to 1,000 mg twice daily between meals
Omega-3, triglyceride form	Leukotriene and inflammatory support	2 to 3 g combined EPA and DHA daily
Vitamin D3 with K2	Immune modulation	Dose to your measured level
Magnesium glycinate	Mast cell membrane stability	300 to 400 mg at bedtime
DAO enzyme	Breaks down dietary histamine before absorption	1 to 2 capsules before high-histamine meals
Multi-strain probiotic	Oral tolerance and immune regulation	25 to 50 billion CFU daily with food
N-acetyl cysteine	Glutathione precursor, mucus thinning	600 mg twice daily away from meals
Colostrum or transfer factor	Mucosal immune support	Per label
Digestive enzymes	Reduces antigenic food particles crossing the barrier	Per label, with meals
Adaptogenic adrenal support	Cortisol rhythm support	Per label, morning and early afternoon

Recommended Lab Tests

Test	Platform	Priority
Food Zoomer	Vibrant Wellness	High: start here for any allergy patient
Wheat Zoomer	Vibrant Wellness	High: rules in or out gluten as a driver
Gut Zoomer	Vibrant Wellness	High: reveals the microbiome and gut barrier picture
Micronutrient Panel	Vibrant Wellness	High: guides targeted supplementation
Total Tox Burden	Vibrant Wellness	High: essential if treatment-resistant or mold history
Organic Acids Test	Vibrant Wellness	Moderate: adds histamine-producing dysbiosis data
Advanced Intestinal Barrier Assessment	Precision Point Diagnostics	High: functional DAO activity; essential when histamine symptoms persist despite dietary changes
AOC1 and HNMT Genetic SNPs	Genova DetoxiGenomic or standalone SNP panel	Moderate to high: explains treatment-resistant histamine intolerance; guides lifelong methylation and cofactor support strategy

> ***Ready to Find Your Root Cause?***
>
> *If you are ready to stop managing your allergies and start looking for what is driving them, the free Root Discovery Session is where we begin.*
>
> *We will review your complete health history, assess your current status through the intake process, and map a clear path forward based on your specific root causes.*
>
> *There is no cost and no obligation.*
>
> *Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.*
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> *Schedule your free Root Discovery Session at rootcauseunlocked.com/discovery, or call the office at (919) 662-0520.*

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